

Example 3

Solve the equation $(x + 3)^2 = 7$.

$$\begin{aligned}\text{If } (x + 3)^2 &= 7 \\ \text{then } x + 3 &= \pm\sqrt{7} \\ x &= -3 \pm \sqrt{7}\end{aligned}$$

Since $\sqrt{7}$ is an irrational number, the roots of the equation $(x + 3)^2 = 7$ are irrational. The roots may be found approximately by writing $\pm\sqrt{7}$ as ± 2.65 .

Exercise 3b

Solve the following equations. If an equation has irrational roots, leave the answer in the form given in Example 3.

- | | |
|--|---|
| ① $(x - 2)^2 = 9$ | ② $(x - 7)^2 = 4$ |
| ③ $(x + 3)^2 = 4$ | ④ $(x + 2)^2 = 25$ |
| ⑤ $(x - 1)^2 = 2$ | ⑥ $(x + 4)^2 = 3$ |
| ⑦ $(x - 3)^2 = 5$ | ⑧ $(x + 2)^2 = 2$ |
| ⑨ $(x - 2)^2 = \frac{1}{4}$ | ⑩ $(x - 6)^2 = 36$ |
| ⑪ $(x - 4)^2 = 10$ | ⑫ $(x + 5)^2 = \frac{1}{9}$ |
| ⑬ $(x + 3)^2 = 49$ | ⑭ $(x - 1)^2 = 7$ |
| ⑮ $(x - 8)^2 = 3$ | ⑯ $(x - 1)^2 = \frac{9}{25}$ |
| ⑰ $(x + 1)^2 = 2\frac{1}{4}$ | ⑲ $(x + 7)^2 = 6$ |
| ⑱ $(x + \frac{1}{3})^2 = \frac{4}{9}$ | ⑳ $(x + 9)^2 = 3$ |
| ㉑ $(x - 6)^2 = 5$ | ㉒ $(x - 2\frac{1}{2})^2 = 6\frac{1}{4}$ |
| ㉓ $(x + 10)^2 = 8$ | ㉔ $(x - 6)^2 = 2\frac{7}{9}$ |
| ㉕ $(x + 1\frac{1}{4})^2 = 1\frac{9}{16}$ | |

3.3 Completing the square

Example 4

What must be added to $x^2 + 6x$ to make the expression a perfect square?

Suppose $x^2 + 6x + k$ is a perfect square and that it is equal to $(x + a)^2$.

$$\begin{aligned}\text{i.e. Let } x^2 + 6x + k &= (x + a)^2 \\ \text{then } x^2 + 6x + k &= x^2 + 2ax + a^2\end{aligned}$$

By comparing coefficients of x ,

$$2a = 6$$

$$a = 3$$

By comparing the constant terms,

$$k = a^2$$

$$k = 3^2 = 9$$

Therefore 9 must be added to the expression.

$$\text{Check: } x^2 + 6x + 9 = (x + 3)^2$$

In practice, the quantity to be added is **the square of half of the coefficient of x** (or whatever letter is involved). In Example 4, the coefficient of x is 6, half of 6 is 3 and the square of 3 is 9. Hence 9 must be added.

Example 5

What must be added to $d^2 - 5d$ to make the expression a perfect square?

Factorise the result.

The coefficient of d is -5 . Half of -5 is $-\frac{5}{2}$.

$$\left(-\frac{5}{2}\right)^2 = +\frac{25}{4}$$

$\frac{25}{4}$ must be added.

$d^2 - 5d + \frac{25}{4}$ is a perfect square.

$$\begin{aligned}d^2 - 5d + \frac{25}{4} &= \left(d - \frac{5}{2}\right)^2 \\ &= \left(d - 2\frac{1}{2}\right)^2\end{aligned}$$

Example 6

Add a term to $n^2 + 1\frac{1}{2}n$ to make the expression a perfect square. Express the result as the square of a bracketed expression.

The coefficient of n is $+1\frac{1}{2}$.

Half of $+1\frac{1}{2} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$ and $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

$\frac{9}{16}$ must be added.

$$n^2 + 1\frac{1}{2}n + \frac{9}{16} = \left(n + \frac{3}{4}\right)^2$$

Examples 5 and 6 can be checked by squaring out the bracket in the final result.

Exercise 3c

In each of the following, add the term that makes the given expression into a perfect square. Write the result as the square of a bracketed expression.

- | | |
|-------------------------|-------------------------|
| ① $a^2 + 8a$ | ② $b^2 + 10b$ |
| ③ $c^2 - 4c$ | ④ $d^2 - 6d$ |
| ⑤ $x^2 + 5x$ | ⑥ $y^2 - 3y$ |
| ⑦ $z^2 - 7z$ | ⑧ $m^2 + 2m$ |
| ⑨ $n^2 - n$ | ⑩ $u^2 - \frac{1}{2}u$ |
| ⑪ $v^2 + \frac{1}{4}v$ | ⑫ $h^2 + \frac{2}{3}h$ |
| ⑬ $k^2 - 1\frac{1}{3}k$ | ⑭ $g^2 - 4\frac{2}{3}g$ |
| ⑮ $a^2 + \frac{3}{5}a$ | ⑯ $b^2 - \frac{4}{5}b$ |
| ⑰ $c^2 - 1\frac{1}{2}c$ | ⑲ $m^2 - 8m$ |
| ⑱ $m^2 - 8mn$ | ⑳ $a^2 - 6ad$ |
| ㉑ $x^2 + 10xy$ | ㉒ $m^2 + 3mn$ |
| ㉓ $u^2 - 1\frac{3}{5}u$ | ㉔ $v^2 - \frac{3}{4}v$ |

Example 7

Solve the equation $x^2 - 8x + 3 = 0$.

The LHS (Left Hand Side) does not factorise. So rearrange the equation to make the LHS a perfect square.

$$x^2 - 8x + 3 = 0$$

Subtract 3 from both sides.

$$x^2 - 8x = -3$$

Add 16 to both sides.

$$x^2 - 8x + 16 = -3 + 16$$

$$\Rightarrow (x - 4)^2 = 13$$

$$\Rightarrow x - 4 = \pm\sqrt{13}$$

$$\Rightarrow x = 4 \pm\sqrt{13}$$

Example 8

Solve the equation $a^2 + 3a - 2 = 0$.

The LHS does not factorise.

$$a^2 + 3a - 2 = 0$$

$$\Rightarrow a^2 + 3a = 2$$

Add the square of $\frac{3}{2}$ to both sides.

$$a^2 + 3a + \left(\frac{3}{2}\right)^2 = 2 + \frac{9}{4}$$

$$= \frac{8 + 9}{4}$$

$$\Rightarrow \left(a + \frac{3}{2}\right)^2 = \frac{17}{4}$$

$$\Rightarrow a + \frac{3}{2} = \pm\sqrt{\frac{17}{4}}$$

$$= \pm\frac{\sqrt{17}}{2}$$

$$\Rightarrow a = -\frac{3}{2} \pm \frac{\sqrt{17}}{2}$$

$$= \frac{-3 \pm \sqrt{17}}{2}$$

If the LHS of the equation factorises, use the method of factorisation rather than completing the square. For example,

$$x^2 - 7x + 10 = 0$$

$$\Rightarrow (x - 5)(x - 2) = 0$$

$$\Rightarrow x = 5 \text{ or } 2$$

rather than

$$x^2 - 7x + 10 = 0$$

$$\Rightarrow x^2 - 7x = -10$$

$$\Rightarrow x^2 - 7x + \left(\frac{7}{2}\right)^2 = -10 + \frac{49}{4}$$

$$= \frac{-40 + 49}{4}$$

$$\Rightarrow \left(x - \frac{7}{2}\right)^2 = \frac{9}{4}$$

$$\Rightarrow x - \frac{7}{2} = \pm\frac{3}{2}$$

$$\Rightarrow x = \frac{7}{2} \pm \frac{3}{2}$$

$$= \frac{10}{2} \text{ or } \frac{4}{2}$$

$$= 5 \text{ or } 2$$

Exercise 3d

Solve the following equations. Factorise where possible. Otherwise, solve by completing the square, leaving the answers in the form of Examples 7 and 8.

- | | |
|-----------------------|------------------------|
| ① $a^2 + 4a - 21 = 0$ | ② $b^2 - b - 12 = 0$ |
| ③ $c^2 - 4c - 2 = 0$ | ④ $d^2 + 2d - 2 = 0$ |
| ⑤ $n^2 + 4n + 4 = 0$ | ⑥ $p^2 - 10p + 15 = 0$ |

- ⑦ $q^2 + 10q + 22 = 0$ ⑧ $t^2 - 6t + 9 = 0$
 ⑨ $m^2 + 6m + 7 = 0$ ⑩ $y^2 - 3y + 1 = 0$
 ⑪ $z^2 - 5z + 6 = 0$ ⑫ $h^2 + 5h + 4 = 0$
 ⑬ $k^2 - 5k + 2 = 0$ ⑭ $g^2 + 5g + 2 = 0$
 ⑮ $x^2 - 8x - 1 = 0$ ⑯ $a^2 - a - 1 = 0$
 ⑰ $b^2 + b - 3 = 0$ ⑱ $y^2 + 7y - 30 = 0$
 ⑲ $m^2 - 7m + 11 = 0$ ⑳ $x^2 + 3x - 2 = 0$
 ㉑ $x^2 - 10x + 25 = 0$ ㉒ $v^2 + 9v + 19 = 0$
 ㉓ $n^2 - 12n + 1 = 0$ ㉔ $u^2 - 14u - 3 = 0$

3.4 The quadratic formula

The **general form of a quadratic equation** is $ax^2 + bx + c = 0$. The roots of the general quadratic equation are found by completing the square.

$$\begin{aligned}
 & ax^2 + bx + c = 0 \\
 \Rightarrow & x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \\
 \Rightarrow & x^2 + \frac{b}{a}x = -\frac{c}{a} \\
 \Rightarrow & x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \\
 & \quad = -\frac{c}{a} + \frac{b^2}{4a^2} \\
 \Rightarrow & \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \\
 \Rightarrow & x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\
 & \quad = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 \Rightarrow & x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
 \Rightarrow & x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
 \end{aligned}$$

This is called the **quadratic formula**. Use the formula to solve *any* quadratic equation.

Example 9

Find, correct to 2 decimal places, the roots of the equation $3x^2 - 5x - 7 = 0$.

Comparing $3x^2 - 5x - 7 = 0$ with

$$ax^2 + bx + c = 0;$$

$$a = 3, b = -5, c = -7.$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 3 \times (-7)}}{2 \times 3}$$

$$= \frac{5 \pm \sqrt{25 + 84}}{6}$$

$$= \frac{5 \pm \sqrt{109}}{6}$$

$$= \frac{5 \pm 10.44}{6}$$

$$= \frac{15.44}{6} \quad \text{or} \quad \frac{-5.44}{6}$$

$$\approx 2.57 \quad \text{or} \quad -0.91$$

Exercise 3e

Use the quadratic formula to solve the following equations. Give the roots correct to 2 decimal places where necessary. Use factorisation to check the results of the first ten.

- ① $x^2 + 5x + 6 = 0$ ② $x^2 - 5x + 4 = 0$
 ③ $x^2 - 4x - 5 = 0$ ④ $2x^2 + 5x + 3 = 0$
 ⑤ $3x^2 - 4x + 1 = 0$ ⑥ $3x^2 - 5x - 2 = 0$
 ⑦ $5x^2 - 3x - 2 = 0$ ⑧ $4x^2 + 7x - 2 = 0$
 ⑨ $6x^2 + 13x + 6 = 0$ ⑩ $3x^2 - 13x - 10 = 0$
 ⑪ $x^2 + 3x + 1 = 0$ ⑫ $x^2 - 2x - 4 = 0$
 ⑬ $2x^2 + 7x - 3 = 0$ ⑭ $3x^2 - 5x - 3 = 0$
 ⑮ $5x^2 - 6x - 3 = 0$ ⑯ $5x^2 + 8x - 2 = 0$
 ⑰ $3x^2 + 7x + 3 = 0$ ⑱ $3x^2 - 12x + 10 = 0$
 ⑲ $3x^2 - 8x + 2 = 0$ ㉑ $5x^2 + 3x - 3 = 0$

3.5 Sum and product of roots

As we have seen earlier, the **roots** of an equation are the values of the unknown that make the equation true. For example, +9 and -1 are the roots of the equation $x^2 - 8x - 9 = 0$.

In this section we show that it is possible to find information about the roots of an equation without solving it. If α and β are the roots of an equation in x , then:

$$\begin{aligned}(x - \alpha)(x - \beta) &= 0 \\ x^2 - \alpha x - \beta x + \alpha\beta &= 0 \\ x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \quad (1)\end{aligned}$$

In equation (1), $(\alpha + \beta)$ is the **sum of the roots** and $\alpha\beta$ is the **product of the roots**. Examples 10 and 11 show how to apply this result.

Example 10

Find the sum and the product of the roots of the equation

a $5x^2 - 4x - 9 = 0$ **b** $2x^2 + 9x = 6$.

a $5x^2 - 4x - 9 = 0$

Dividing throughout by 5

$$x^2 - \frac{4}{5}x - \frac{9}{5} = 0$$

Comparing this with equation (1)

$$(\alpha + \beta) = \frac{4}{5} \text{ and } \alpha\beta = -\frac{9}{5}$$

The sum of the roots is $\frac{4}{5}$ and their product is $-\frac{9}{5}$.

b $2x^2 + 9x = 6$

$$2x^2 + 9x - 6 = 0$$

Dividing throughout by 2

$$x^2 + \frac{9}{2}x - 3 = 0$$

Comparing this with equation (1)

$$(\alpha + \beta) = -\frac{9}{2} \text{ and } \alpha\beta = -3$$

The sum of the roots is $-\frac{9}{2}$ and their product is -3 .

Example 11

Find the equation whose sum and product of roots are respectively **a** $-5, 6$, **b** $1.5, -1.2$.

Write each answer in the form $ax^2 + bx + c = 0$.

Let the sum and product of roots be $(\alpha + \beta)$ and $\alpha\beta$ as in equation (1) above, then

a $(\alpha + \beta) = -5$ and $\alpha\beta = 6$

Substitute these values into

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

to give:

$$x^2 - (-5)x + 6 = 0$$

$$x^2 + 5x + 6 = 0$$

b $(\alpha + \beta) = 1.5$ and $\alpha\beta = -1.2$

Substitute these values into

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

to give:

$$x^2 - (1.5)x - 1.2 = 0$$

$$x^2 - 1.5x - 1.2 = 0$$

$$x^2 - \frac{3}{2}x - \frac{6}{5} = 0$$

Multiply by 10 throughout:

$$10x^2 - 15x + 12 = 0$$

Exercise 3f

① Use the method of Example 10 to find the sums and products of the roots of the following equations.

a $2x^2 - 10x + 7 = 0$

b $3x^2 + x - 5 = 0$

c $4x^2 = 3x + 1$

d $y^2 + 5y = 1$

e $a(a - 1) = 5$

f $x + \frac{1}{x} = 3$

② Use the method of Example 11 to find equations in the form $ax^2 + bx + c = 0$ whose sums and products of roots are respectively:

a $3, 4$

b $-5, 6$

c $\frac{3}{2}, -\frac{5}{2}$

d $-\frac{7}{3}, 0$

e $0, -7$

f $1.2, 0.8$

3.6 Word problems leading to quadratic equations

Example 12

Find two numbers whose difference is 5 and whose product is 266.

Let the smaller number be x .
Then the larger number is $x + 5$.
Their product is $x(x + 5)$.

$$\begin{aligned}\text{Hence } x(x + 5) &= 266 \\ \Rightarrow x^2 + 5x - 266 &= 0 \\ \Rightarrow (x - 14)(x + 19) &= 0 \\ \Rightarrow x &= 14 \text{ or } -19\end{aligned}$$

The other number is $14 + 5$ or $-19 + 5$
i.e. 19 or -14 .

The two numbers are 14 and 19, or -19 and -14 .

Check: $14 \times 19 = 266$ and $-19 \times -14 = 266$

Example 13

A woman is 4 times older than her child. 5 years ago the product of their ages was 175. Find their present ages.

Let the child's age be x years.
Then the mother's age is $4x$ years.
5 years ago, the child's age was $(x - 5)$ years
and the mother's age was $(4x - 5)$ years.
The product of their ages was $(x - 5)(4x - 5)$.

$$\begin{aligned}\text{Hence } (x - 5)(4x - 5) &= 175 \\ \Rightarrow 4x^2 - 25x + 25 &= 175 \\ \Rightarrow 4x^2 - 25x - 150 &= 0 \\ \Rightarrow (4x + 15)(x - 10) &= 0 \\ \Rightarrow x &= 10 \text{ or } -\frac{15}{4}\end{aligned}$$

However, $-\frac{15}{4}$ is not sensible for an age (ages cannot be negative). Therefore the child is 10 years old and the mother is 40 years old.

Check: $(40 - 5)(10 - 5) = 35 \times 5 = 175$

Notice, in Example 13, that one of the roots does not give a sensible answer. Disregard any root which is not sensible in terms of the given question.

Exercise 3g

- ① Find two numbers which differ by 4 and whose product is 45.
- ② The width of a classroom is 4 m less than the length. Its area is 45 m^2 . Find the dimensions of the classroom.
- ③ Two numbers have a difference of 3. The sum of their squares is 89. Find the numbers.
- ④ Two square rooms have a total floor area of 89 m^2 . One room is 3 m longer each way than the other. Find the dimensions of the two rooms.
- ⑤ A girl is 6 years younger than her brother. The product of their ages is 135. Find their ages.
- ⑥ The ages of two children are 11 and 8 years. In how many years' time will the product of their ages be 208?
- ⑦ Find the number which, when added to its square, makes 90.
- ⑧ Twice the square of a certain whole number added to 3 times the number makes 90. Find the number.
- ⑨ The area of a rectangle is 60 cm^2 . The length is 11 cm more than the width. Find the width.
- ⑩ A rectangular plot measures 12 m by 5 m. A path of constant width runs along one side and one end. If the total area of the plot and the path is 120 m^2 , find the width of the path. (Hint: Make a sketch and let the width of the path be $x \text{ m}$.)
- ⑪ A rectangular piece of cardboard measures 17 cm by 14 cm. Strips of equal width are cut off one side and one end. The area of the remaining piece is 108 cm^2 . Find the width of the strips removed.
- ⑫ A man is 37 years old and his child's age is 8. How many years ago was the product of their ages 96?

- 13 A certain number is subtracted from 18 and from 13. The product of the two numbers obtained is 66. Find the first number.
- 14 Find two consecutive numbers whose product is 156.
- 15 Find two consecutive even numbers whose product is 224.
- 16 Find two consecutive odd numbers whose product is 195.
- 17 The square of a certain number is 22 less than 13 times the original number. Find the number.
- 18 A woman is 3 times as old as her son. 8 years ago the product of their ages was 112. Find their present ages.
- 19 Twice a certain whole number subtracted from 3 times the square of the number leaves 133. Find the number.
- 20 The base of a triangle is 3 cm longer than its corresponding height. If the area is 44 cm^2 , find the length of its base.

SUMMARY

- To solve a quadratic equation means to find its roots. To find the roots:
 - arrange the equation in the form $ax^2 + bx + c = 0$
 - factorise the LHS if possible
 - use the result that if $pq = 0$, then $p = 0$ or $q = 0$ (or both $= 0$) to find the roots. See Examples 1 and 2.
- If a quadratic expression $ax^2 + bx + c = 0$ does *not* factorise, solve the equation either:
 - by completing the square (as in Examples 4–8), or
 - by applying the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
 See Example 9.
- It is possible to form a quadratic equation, given the *sum* and *product* of its roots. See Example 11.
- Some *word problems* can be expressed as quadratic equations. The equation can then be used to solve the problem.

Challenge Box

Switching Units and Tens

Look at these multiplications:

$$32 \times 46 = 1\,472$$

$$23 \times 64 = 1\,472$$

Check that they are correct and explain why this works.

Can you make up others like this?

Challenge Box

Infinity and Beyond

In this series, each number is $\frac{1}{4}$ of the previous number:

$$\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \frac{1}{4^4} + \dots$$

What do you get if you add these numbers together?

What if you extend the series by more terms ... until you get to ... **infinity**?

Is a calculator any help?

Is there a smarter way?

4

Numerical processes (2) Approximation and errors

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Round off given values to the nearest ten, hundred, thousand, etc., or to a given number of decimal places and/or significant figures
- 2 Use rounded values to estimate the outcome of calculations
- 3 Calculate the percentage error when using rounded values
- 4 Decide on the degree of accuracy that is appropriate to given data which may have been rounded.

Teaching and learning materials

Teacher: Newspaper articles and reports that contain numerical data, population and other official data.

Student: 4-figure tables (provided on pages 237 to 248 of this book), calculator.

4.1 Approximations

Approximation is the process of using rounded values when presenting numerical data and making rough calculations. It also involves the analysis of outcomes arising from using rounded values.

Rounding off (revision)

Numbers can be rounded off to the nearest hundred, ten, whole number, etc., or to a given number of decimal places (d.p.).

The digits 1, 2, 3 and 4 are rounded down and the digits 5, 6, 7, 8 and 9 are rounded up.

Example 1

Round off the number 163.864

- a* to 2 decimal places,
- b* to 1 decimal place,
- c* to the nearest whole number,
- d* to the nearest hundred.

$$a \quad 163.864 = 163.86 \quad \text{to 2 d.p.}$$

$$b \quad 163.864 = 163.9 \quad \text{to 1 d.p.}$$

$$c \quad 163.864 = 164 \quad \text{to the nearest whole number}$$

$$d \quad 163.864 = 200 \quad \text{to the nearest hundred}$$

Significant figures (revision)

Numbers are sometimes rounded off to a given number of significant figures (s.f.). The significance of a digit depends on its position in the number. Thus in the number 146.83, the 4 is more significant than the 6, the 6 is more significant than the 8 and the 1 is the most significant digit.

The first significant figure in a decimal fraction is the first non-zero digit in the fraction. For example, the first significant figure in the decimal 0.002 487 is 2.

Example 2

Round off 146.83

- a* to 1 significant figure,
- b* to 2 significant figures,
- c* to 3 significant figures,
- d* to 4 significant figures.

$$a \quad 146.83 = 100 \text{ to 1 s.f.}$$

$$b \quad 146.83 = 150 \text{ to 2 s.f.}$$

$$c \quad 146.83 = 147 \text{ to 3 s.f.}$$

$$d \quad 146.83 = 146.8 \text{ to 4 s.f.}$$

Example 3

Round off 0.002 487 to

- a 1 s.f., b 2 s.f., c 3 s.f.

a 0.002 to 1 s.f.

b 0.0025 to 2 s.f.

c 0.00249 to 3 s.f.

Example 4

Round off

a 8.026 to 3 s.f., b 50.95 to 2 s.f.,

c 18.057 to 1 decimal place,

d $\text{N}450\,170$ to 3 s.f.

a $8.026 = 8.03$ to 3 s.f.

b $50.95 = 51$ to 2 s.f.

c $18.057 = 18.1$ to 1 d.p.

d $\text{N}450\,170 = \text{N}450\,000$ to 3 s.f.

Example 5

In one year Nigeria's food imports cost $\text{N}13$ billion to 2 s.f. What is the range of values of the cost?

The range of values is between $\text{N}12.5$ billion and $\text{N}13.5$ billion.

Estimation

Approximations are very useful when making **rough estimates** of calculations. For example if a person pays $\text{N}3\,655$ per month towards paying for an iphone, then the total amount for a year is $\text{N}3\,655 \times 12$. Using values correct to 1 s.f., a rough estimate of this is:

$$\text{N}4\,000 \times 10 = \text{N}40\,000.$$

Rough estimates are not absolutely accurate (for example the true amount for the iphone is $\text{N}43\,860$). However, they give an idea of the **order of magnitude** of the outcome of a calculation. Thus they are often used to check the correctness of answers to calculations. See their use in Chapter 1.

Example 6

Use a calculator to find the value of

$$\frac{1.386 \times 0.0894}{15.67}.$$

Check your result by making a rough estimate.

$$\frac{1.386 \times 0.0894}{15.67} = \frac{0.123\,9084}{15.67} = 0.007\,9074 \quad (\text{calculator})$$

$$\text{Rough estimate: } \frac{1 \times 0.09}{20} = \frac{0.09}{20} = 0.0045$$

In Example **b** the answer and the rough estimate are of the same order of magnitude. If the rough estimate and answer are not of the same order, then they should be rechecked. *Note:* the numbers in Example 6 were rounded to one significant figure (1 s.f.).

Exercise 4a

- ① Round off the following to the nearest whole number.

a 3.24 m

b 0.97 m

c 86.02 ha

d 341.77 cm

e 496 km

f 164.90 mm

g 52.84

h $\$18.07$

i 129.7 litres

j 9.63

- ② Round off the following to 1 decimal place.

a 0.0786

b 18.04

c 18.954

d 0.786

e 7.926

f 20.08

- ③ Round off the following to 2 significant figures.

a 36.9

b 109.4

c $0.009\,281$

d 4.98

e $0.024\,39$

f $86\,125$

g 144

h 9.04

i 2.99

j 304.9

- ④ Round off the following to 3 significant figures.

a $7\,579$

b $52\,069$

c $352\,289$

d $1\,789$

e $0.089\,21$

f 170.08

g 83.352

h 0.9064

i $827\,502$

j 80.96

- ⑧ Round off the following to given degrees of accuracy:

- a 2 748 924 000 to the nearest billion
- b 2 748 924 000 to the nearest million
- c ₦482 677 to the nearest ₦10 000
- d 923 768 km² to the nearest 1 000 km²
- e 923 768 km² to the nearest million km²

- ⑨ In 2008, Nigeria's rate of oil production was 1.825 million barrels a day.

- a Give at least three ways in which a newspaper might have reported this amount.
- b Use estimation to decide whether or not oil production reached 1 billion barrels in 2008.

- ⑦ (oral) Table 4.1 shows the estimated populations of five states in 2008.

State	Population
Kwara	2 591 600
Lagos	17 552 900
Oyo	6 617 700
Rivers	6 689 100
Sokoto	4 244 400

Table 4.1

Round off the populations to

- a 1 s.f., b 2 s.f., c the nearest thousand.

- ⑧ A newspaper headline reads, 'Government rejects US\$23 billion loan from IMF'. Between what two amounts does this figure lie?

- ⑨ Table 4.2 shows the numbers of people departing from an international airport for the years 2007 to 2011.

2007	2008	2009	2010	2011
320 120	269 546	532 705	501 992	646 155

Table 4.2

- a Round off the numbers to the nearest hundred thousand.
- b Use your values to estimate how many million people departed from the airport during the five years.

- ⑩ In 2010 Nigeria's estimated reserves of foreign exchange and gold amounted to US\$49.67 billion and the rate of exchange was ₦150.48 to 1US\$. You are to estimate the amount of reserves in naira.

- a What would be sensible rounded values to use?
- b Use these values to find the reserves in trillions of naira (nearest trillion).

4.2 Percentage error

No measurement, however carefully made, is exact. Measurements are approximate. If the width of a room is measured as 2.8 m to 2 s.f., the actual width may lie between 2.75 m and 2.85 m. The error of this measurement is $2.75 - 2.8$ m or $2.85 - 2.8$ m, that is -0.05 m or $+0.05$ m.

Fig. 4.1 shows the range of actual measurement.

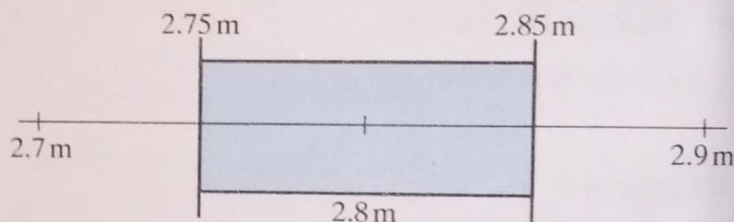


Fig. 4.1

$$\text{Percentage error} = \frac{\text{error}}{\text{measurement}} \times 100\%$$

In Fig. 4.1,

$$\begin{aligned} \text{percentage error} &= \pm \frac{0.05}{2.8} \times 100\% \\ &= \pm 1.78571\% \text{ (calculator)} \\ &= \pm 1.8\% \text{ (to 2 s.f.)} \end{aligned}$$

Suppose the width of the same room is measured, to the nearest centimetre, as 2 m 80 cm. The range of the actual measurement is between 2 m 79.5 cm and 2 m 80.5 cm.

$$\text{Error} = \pm 0.5 \text{ cm}$$

$$\begin{aligned} \text{Percentage error} &= \frac{\pm 0.5 \text{ cm}}{280 \text{ cm}} \times 100\% \\ &= \pm 0.17857\% \text{ (calculator)} \\ &= \pm 0.18\% \text{ to 2 s.f.} \end{aligned}$$

From the above, it can be seen that the percentage error is smaller when the centimetre is used for measurement. Thus, when measuring, the degree of accuracy improves as the unit of measurement decreases in size.

Example 7

The length of a pole is measured as 5 metres to the nearest metre. What is the range of its actual length? Calculate the percentage error.

Since the measurement is given to the nearest metre, the actual length of the pole will be in the range 4.5 m to 5.5 m.

$$\text{Error} = \pm 0.5 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \pm \frac{0.5 \text{ m}}{5 \text{ m}} \times 100\% \\ &= \pm 10\%\end{aligned}$$

Example 8

The length of a running track is measured and given as 400 m. Find the percentage error if the length is measured **a** to the nearest metre, **b** to the nearest 10 m, **c** to 1 significant figure.

- a** The range of actual measurement is between 399.5 m and 400.5 m.

$$\text{Error} = \pm 0.5 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 0.5 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 0.125\%\end{aligned}$$

- b** The range of actual measurement is between 395 m and 405 m.

$$\text{Error} = \pm 5 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 5 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 1\frac{1}{4}\%\end{aligned}$$

- c** The range of actual measurement is between 350 m and 450 m.

$$\text{Error} = \pm 50 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 50 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 12\frac{1}{2}\%\end{aligned}$$

Example 9

A stick of length 1.75 m was measured by a boy as 1.80 m. Find the percentage error in his measurement. [WAEC]

$$\begin{aligned}\text{Error in measurement} &= 1.8 - 1.75 \text{ m} \\ &= +0.05 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\text{error}}{\text{true value}} \times 100\% \\ &= +\frac{0.05}{1.75} \times 100\% \\ &= +\frac{5}{1.75}\% \\ &= +\frac{20}{7}\% \\ &= +2\frac{6}{7}\%\end{aligned}$$

Example 10

A square is 5 cm by 5 cm. A student measures a side of the square as 4.9 cm. She uses her measurement to calculate the area of the square. Find the percentage error in **a** the length of the side, **b** the area of the square.

a Error in measurement = 4.9 – 5 cm
= –0.1 cm

$$\begin{aligned}\text{Percentage error} &= -\frac{0.1}{5} \times 100\% \\ &= -2\%\end{aligned}$$

b Actual area of square = $5 \times 5 \text{ cm}^2 = 25 \text{ cm}^2$
Calculated area = $4.9 \times 4.9 \text{ cm}^2$
= 24.01 cm^2
(from tables)

$$\begin{aligned}\text{Error in area} &= 24.01 - 25 \text{ cm}^2 \\ &= -0.99 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Percentage error} &= -\frac{0.99}{25} \times 100\% \\ &= -0.99 \times 4\% \\ &= -3.96\%\end{aligned}$$

Note: in questions such as Examples 9 and 10, it is quite common to ignore the sign of the percentage error. Thus answers of $2\frac{6}{7}\%$, 2% and 3.96% are acceptable in these questions.

Exercise 4b

- ① What is the range of values of each of the following measurements?
 - a** The length of a line segment is 9 cm to the nearest centimetre.
 - b** A person is 1.75 m tall to the nearest centimetre.

- c The radius of the earth is 6400 km to 2 s.f.
d The distance from the school to the market is 9.8 km to 1 d.p.
e The population of Nigeria is 150 million to the nearest 10 million.
f A container holds 7.5 litres of petrol to the nearest 0.1 litre.
g A person spent ₦20 000 to the nearest thousand naira.
h The mass of a wrestler is 75.6 kg to the nearest 0.1 kg.
i The maximum temperature for a particular day in Lagos is 30.2°C to the nearest 0.1°C .
j The volume of a sphere is 860 cm^3 to the nearest cm^3 .
- 2 Calculate the percentage error in the measurements in question 1.
- 3 Calculate the percentage error in each of the following measurements.
- The capacity of a bucket is 7.5 litres to 1 d.p.
 - The distance between two towns is 60 km to the nearest kilometre.
 - A pole is 125 cm high to the nearest centimetre.
 - The volume of a box is 25 cm^3 to the nearest cm^3 .
 - The thickness of a book is 20 mm to the nearest millimetre.
 - The mass of a student is 62 kg to 2 s.f.
 - The speed of an aircraft is 800 km/h to 1 s.f.
 - The radius of a ball is 21 cm to the nearest centimetre.
 - The area of a classroom is 400 m^2 to 2 s.f.
 - The area of Nigeria is $900\,000\text{ km}^2$ to 1 s.f.
- 4 The length of a stick is 8 cm. A student measures the length as 8.5 cm. Find the percentage error in the measurement.
- 5 A pencil is 18 cm long. Someone estimates its length to be 20 cm. Find the percentage error in the estimate.
- 6 A student draws a line and says that it is 10 cm long. When it is carefully measured, the true length is 10.2 cm. What is the percentage error in the drawing?
- 7 A can of cola says that it contains 330 ml. A food inspector carefully measures the contents of a can and finds that it contains 341 ml. Calculate the percentage error in the contents.
- 8 A stick is 20 cm long. A student makes a 5% error in measuring the stick. Find two possible values for the student's measurement.
- 9 A surveyor measures a road as being 69.3 km long, however, there is a -1% error in this measurement. What is the true length of the road?
- 10 A square is 10 cm by 10 cm. A student measures a side of the square as 9.9 cm and uses the measurement to calculate the area of the square. Find the percentage error in
- the length of the side,
 - the area of the square.

4.3 Degree of accuracy

Many calculations involve given values, often measurements. The degree of accuracy of such values affects the degree of accuracy of the results of the calculations. Therefore, the degree of accuracy of the values in a calculation must be taken into consideration when determining the answer to the calculation. Normally, the final result of a calculation should not be given to a number of significant figures more than the number of significant figures in any of the given data.

Rounded-off values are sometimes used in calculations. For example, π is often taken as 3.14 or 3.142.

Sometimes it seems that an answer found at an intermediate stage of a calculation may need to be rounded off before it is used in a subsequent stage. However, since such rounded-off values affect the degree of accuracy of the results of calculations, it is generally not advisable to round off intermediate values.